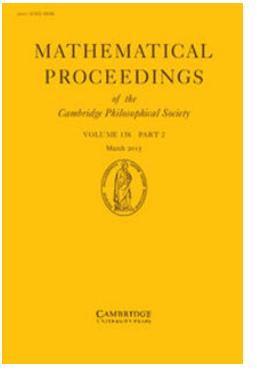
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On the zeros of the Riemann zeta-function

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*On the zeros of the Riemann zeta-function**. By Mr J. E. LITTLEWOOD, Trinity College, Cayley Lecturer.

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1. Let $N(T)$ denote, as usual, the number of zeros of $\zeta(s)$ whose imaginary part γ satisfies $0 < \gamma < T$, and $N(\sigma, T)$ the number of these for which, in addition, the real part is greater than σ. In this definition we suppose, in the first place, that no zero actually lies on the line $t = T$: if the line contains zeros we define

$$N(T) = \lim_{\epsilon \to 0} \tfrac{1}{2} \{N(T+\epsilon) + N(T-\epsilon)\},$$

$$N(\sigma, T) = \lim_{\epsilon \to 0} \tfrac{1}{2} \{N(\sigma, T+\epsilon) + N(\sigma, T-\epsilon)\}\dagger.$$

Zeros on a boundary $t = constant$ are, in fact, only to count half the full value‡.

We define a function $f(s) = \log \zeta(s)$ as follows. In the first place when $\sigma > 1$,

$$f(s) = \sum_{m,p} \frac{1}{m} p^{-ms},$$

the familiar Dirichlet series. Next, if no zero of $\zeta(s)$ has ordinate t, we define $f(\sigma + it)$ to be the value obtained from $f(2 + it)$ by continuous variation of the argument along the straight line from $2 + it$ to $\sigma + it$. Finally, for a t which is the ordinate of a zero we define

$$f(\sigma + it) = \lim_{\epsilon \to 0} \tfrac{1}{2} \{f(\sigma + it + i\epsilon) + f(\sigma + it - i\epsilon)\}.$$

It is known§ that, as $T \to \infty$,

$$N(T) = M(T) + S(T) + \tfrac{7}{8} + O(T^{-1}) \tag{1.1}$$

$$= M(T) + S(T) + O(1),$$

where

$$M(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi}, \tag{1.2}$$

$$S(T) = \frac{1}{\pi} \operatorname{I\!I} f(\tfrac{1}{2} + iT). \tag{1.3}$$

It is known further that

$$\operatorname{I\!I} f(s) = O(\log t) \quad (\sigma \geqslant \tfrac{1}{2}), \tag{1.4}$$

* Most of the results of the paper were announced, without proof, in the *Records of Proceedings of the London Mathematical Society*, February 10, 1921.

† These equations are, of course, then valid for *all* T.

‡ We adhere strictly, however, to the condition $\beta > \sigma$ for the real part of a zero counted in $N(\sigma, T)$. A halving convention here has something against it and nothing in its favour.

§ R. Backlund, *Acta Mathematica*, 41 (1918), 345–375. With our conventions the results are evidently true for all T if true for "*wurzelfrei*" T.

and in particular

$$S(T) = O(\log t). \tag{1.5}$$

The present paper is devoted to the study of the functions $N(\sigma, T)$ and $S(T)$: it falls naturally into two parts. In Theorem 2 we prove that

$$\int_\sigma^1 N(\sigma, T)\, d\sigma \text{ and } \frac{1}{2\pi}\int_0^T \log|\zeta(\sigma + it)|\, dt$$

on the one hand, and in Theorem 3 that

$$\int_0^T \mathbb{R}f(\sigma + it)\, dt \text{ and } \int_\sigma^\infty \log|\zeta(\sigma + iT)|\, d\sigma$$

on the other, are nearly equal when $\sigma \geqslant \frac{1}{2}$. From the first result we deduce upper bounds for $N(\sigma, T)$ when σ is near the value $\frac{1}{2}$; and from the second it follows eventually that

$$\int_0^T S(t)\, dt = O(\log T),$$

an upper bound which may be improved to $o(\log T)$ if we assume the hypothesis of Lindelöf (*a fortiori* if we assume Riemann's hypothesis). These are the principal results of the first part. In the second we assume the truth of the Riemann hypothesis, and prove, among other things, that

$$S(T) = O\left(\frac{\log T}{\log\log T}\right).$$

Throughout the paper A's and a's denote positive absolute constants, not always the same from one occurrence to another*. On the other hand, an A or a affected with a suffix is the same wherever it occurs. O's are absolute unless the contrary is stated†.

When we are concerned with approximations valid when $T \to \infty$ we need only consider values of T greater than any convenient constant. We shall suppose in such cases, without always making an explicit statement, that T satisfies any inequality $T > A$ which is required by the run of the argument.

2. We begin by collecting some results whose proof would interrupt the development of later arguments.

Lemma 1.

$$\int_2^T |\zeta(\sigma + it)|^2\, dt < AVT \quad (T \geqslant 2, \tfrac{1}{2} \leqslant \sigma \leqslant 2),$$

where

$$V = \operatorname{Min}\left(\log T, \frac{1}{\sigma - \frac{1}{2}}\right).$$

* Thus we write, e.g. $(\sin\theta + 2)^3 < (A + A)^a < A$.

† $f(x) = O\{\phi(x)\}$ means $|f(x)/\phi(x)| < K$ $(x > x_0)$: an O involves two constants, K and x_0. Our convention is that both are A's.

The lemma is closely allied to known results*, but it does not actually follow from any published theorem, and a proof must be given *de novo*. We have†, for $\frac{1}{2} \leqslant \sigma \leqslant 2$, $t \geqslant 2$,

$$|\zeta(s)| = \left|\sum_{n<t} n^{-s} + O(1)\right| < \left|\sum_{n<t}\right| + A,$$

$$\int_2^T |\zeta(\sigma+it)|^2\,dt \leqslant 2\int_2^T \left|\sum_{n<t} n^{-s}\right|^2 dt + 2A. \tag{2.1}$$

Now

$$\int_2^T \left|\sum_{n<t} n^{-s}\right|^2 dt = \int_2^T \sum_{m,n<t} (mn)^{-\sigma}\left(\frac{m}{n}\right)^{it} dt = \sum_{m,n<T} \int_{T_1}^T (mn)^{-\sigma}\left(\frac{m}{n}\right)^{it} dt,$$

where $T_1 = \text{Max}(m, n, 2)$,

$$\leqslant \sum_{m=n} \int_{T_1}^T + \sum_{m\neq n} \left|\int_{T_1}^T\right|$$

$$\leqslant \sum_{m<T} Tm^{-2\sigma} + 2\sum'_{m,n<T} (mn)^{-\sigma} \Big/ \left|\log\frac{n}{m}\right| = S_1 + S_2 \tag{2.2}$$

say, where the dash denotes that terms for which $m = n$ are to be omitted in Σ'. Now

$$S_1 \leqslant T\,\text{Min}\left(\sum_{m<T} m^{-1}, \sum_1^\infty m^{-2\sigma}\right) \tag{2.3}$$

$$< AT\,\text{Min}\left(\log T, \frac{1}{\sigma-\frac{1}{2}}\right) = AVT.$$

Also if $n = m + r$ we have

$$n^{-\sigma} \Big/ \left|\log\frac{n}{m}\right| < \begin{cases} Am^{-\sigma} \Big/ \dfrac{|r|}{m} & (|r| < \frac{1}{2}m) \\ An^{-\sigma} & (|r| \geqslant \frac{1}{2}m). \end{cases}$$

Hence

$$S_2 < \sum_{m<T} m^{-\sigma}\left(m^{1-\sigma} A \sum_{r<\frac{1}{2}m} \frac{1}{r} + A\sum_{n<T} n^{-\sigma}\right)$$

$$< A\sum_{m<T} m^{1-2\sigma}\log(m+1) + A\left(\sum_{m<T} m^{-\sigma}\right)^2$$

$$< A\int_1^T u^{1-2\sigma}\log(u+1)\,du + A\log T + A\left(\sum_{m<T} m^{-\frac{1}{2}}\right)^2$$

If $\sigma > \frac{3}{4}$,

$$S_2 < A\int_1^T u^{-\frac{1}{2}} Au^{\frac{1}{2}}\,du + AT + AT < AT < AVT,$$

* See G. H. Hardy and J. E. Littlewood, *Proc. London Math. Soc.* (2), 21 (1921), 39–74.

† *Loc. cit.*, 53.

since $V \geqslant \dfrac{1}{\sigma - \frac{1}{2}} \geqslant \dfrac{1}{2 - \frac{1}{2}} = A$ when T is large. If $\sigma \leqslant \frac{3}{4}$

$$S_2 < A\left[\frac{u^{2-2\sigma}\log(u+1)}{2-2\sigma}\right]_1^T - \int_1^T \frac{u^{2-2\sigma}}{2-2\sigma}\frac{du}{u+1} + AT$$

$$< \frac{AT^{2-2\sigma}\log(T+1)}{2-\frac{3}{2}} + AT < AT\log T \,.\, T^{-2(\sigma-\frac{1}{2})} + AT.$$

The first term does not exceed $AT\log T$ on the one hand, and is equal to

$$\frac{T}{2(\sigma-\frac{1}{2})}xe^{-x} \leqslant \frac{T}{2(\sigma-\frac{1}{2})} \quad (x = (2\sigma-1)\log T)$$

on the other. Thus

$$S_2 < AVT + AT < AVT$$

also in this case, and so for $\frac{1}{2} \leqslant \sigma \leqslant 2$. From this and (2.3) the lemma follows.

3. *Lemma* 2. *Suppose that z_1 and z_2 lie within $|z| \leqslant 4$, and that $z_2 - z_1$ is real, positive, and not greater than η. Then*

$$\left|\Re\int_{z_1}^{z_2}\log z\,dz\right| < A\eta\log\left(\frac{1}{\eta}+2\right),$$

the integral being taken along a straight line.

This is, of course, nearly trivial. Suppose first that $|z_1| > 2\eta$. Then $|z| > \eta$ for any point of the path of integration, and so

$$|\Re\log z| \leqslant \left|\log\frac{1}{\eta}\right| + \log 4,$$

$$\left|\Re\int_{z_1}^{z_2}\log z\,dz\right| \leqslant \eta\left(\left|\log\frac{1}{\eta}\right| + \log 4\right) < A\eta\log\left(\frac{1}{\eta}+2\right).$$

Suppose now $|z_1| \leqslant 2\eta$. Then $|z_2| \leqslant 3\eta$. Also, $\log z$ being interpreted to have its principal value, we have, writing

$$r_1 = |z_1|, \quad r_2 = |z_2|,$$

(3.1)

$$\left|\int_{z_1}^{z_2}\log z\,dz\right| \leqslant \left|\int_0^{z_1}\right| + \left|\int_0^{z_2}\right| \leqslant |z_1\log z_1| + |z_1| + |z_2\log z_2| + |z_2|$$

$$\leqslant r_1|\log r_1| + (\pi+1)r_1 + r_2|\log r_2| + (\pi+1)r_2.$$

Since $r|\log r|$ increases to a maximum at $r = e^{-1}$, then decreases to a minimum at $r = 1$, and then increases, we have

$$r_1|\log r_1| \leqslant \begin{cases} 3\eta\log\dfrac{1}{3\eta} & (3\eta < e^{-1}) \\ A & (3\eta > e^{-1}), \end{cases}$$

and in either case

$$r_1 \mid \log r_1 \mid + Ar_1 < A\eta \log \left(\frac{1}{\eta} + 2\right).$$

Since a similar result holds for r_2 (3.1) leads to the desired conclusion also when $\mid z_1 \mid \leqslant 2\eta$.

Lemma 3. *Let C_1, C_2, C_3 be three simple closed curves, C_3 containing C_2, and C_2 containing C_1. Suppose that $f(z)$ is regular and satisfies $\mid f \mid \leqslant M_3$ in the interior of C_3, and satisfies $\mid f \mid \leqslant M_1$ on C_1. Then within and upon C_2*

$$\mid f \mid \leqslant M_1^{\alpha} M_3^{\beta},$$

where α, β are positive numbers satisfying $\alpha + \beta = 1$, and depending only on the geometrical configuration of the set of curves (not on its absolute position or orientation).

This generalization by conformal representation of Hadamard's "three circles" theorem is a known result*.

4. Suppose $\phi(s)$ is meromorphic in and upon the boundary of a rectangle bounded by the lines $t = 0$, $t = T$; $\sigma = \alpha$, $\sigma = \beta$ $(\beta > \alpha)$; and regular and never zero on the side, l, of the rectangle for which $\sigma = \beta$. The function $\log \phi(s)$ is regular in a neighbourhood containing l, and we define

$$F(s) = \log \phi(s)$$

for s of l. For other s of the rectangle we define $F(s)$ to be the value obtained from $\log \phi(\beta + it)$ by continuous variation along $t = constant$ from $\beta + it$ to $\sigma + it$. This provided the path does not cross a zero or pole of $\phi(s)$; if it does we define

$$F(s) = \tfrac{1}{2} \lim \{F(\sigma + it + i\epsilon) + F(\sigma + it - i\epsilon)\}.$$

Let now $\nu(\sigma', T)$ denote the excess of the number of zeros over the number of poles in the part of the rectangle for which $\sigma > \sigma'$, zeros or poles on $t = 0$ or $t = T$ counting one-half only. Then we have

Theorem 1. $$\int F(s)\, ds = -2\pi i \int_{\alpha}^{\beta} \nu(\sigma, T)\, d\sigma,$$

the integral on the left being taken round the rectangle in the positive direction.

This formula has obvious affinities with Jensen's well-known theorem. Various proofs are possible; the one given here depends on the inversion of the order of a double integration. We may suppose $t = 0$ and $t = T$ to be free of zeros and poles of $\phi(s)$; it is

* H. Bohr, E. Landau, and J. E. Littlewood, *Bull. Acad. Belgique*, 15 (1913), 1144–1175, 1166.

easily verified that our halving conventions then ensure the truth of the theorem in the general case. We have

$$\begin{aligned}\int F(s)\,ds &= \int_\alpha^\beta F(\sigma)\,d\sigma - \int_\alpha^\beta F(\sigma+iT)\,d\sigma + \int_0^T \{F(\beta+it) - F(\alpha+it)\}\,idt \\ &= \int_\alpha^\beta F(\sigma)\,d\sigma - \int_\alpha^\beta F(\sigma+iT)\,d\sigma + \int_0^T idt \int_\alpha^\beta \frac{\phi'}{\phi}(\sigma+it)\,d\sigma^* \\ &= \int_\alpha^\beta F(\sigma)\,d\sigma - \int_\alpha^\beta F(\sigma+iT)\,d\sigma + \int_\alpha^\beta d\sigma \int_\sigma^{\sigma+iT} \frac{\phi'}{\phi}(s)\,ds.\end{aligned}$$

Now by the theorem of residues the (inner) integral last written is equal to

$$\int_\sigma^\beta \frac{\phi'}{\phi}(\sigma)\,d\sigma + \int_0^T \frac{\phi'}{\phi}(\beta+it)\,idt - \int_\sigma^\beta \frac{\phi'}{\phi}(\sigma+iT)\,d\sigma - 2\pi i\nu(\sigma, T).$$

Hence

$$\begin{aligned}\int F(s)\,ds &= \int_\alpha^\beta \{F(\sigma) - F(\sigma+iT) + F(\beta) - F(\sigma) + F(\beta+iT) \\ &\qquad - F(\beta) - F(\beta+iT) + F(\sigma+iT) - 2\pi i\nu(\sigma, T)\}\,d\sigma \\ &= -2\pi i \int_\alpha^\beta \nu(\sigma, T)\,d\sigma.\end{aligned}$$

5. *Theorem* 2.

(5.1)

$$2\pi \int_\sigma^1 N(\sigma, T)\,d\sigma = \int_0^T \log|\zeta(\sigma+it)|\,dt + c(\sigma) + \mathrm{I}\int_\sigma^\infty f(\sigma+iT)\,d\sigma$$

(5.2)
$$= \int_0^T \log|\zeta(\sigma+it)|\,dt + O(\log T) \quad (-1 \leqslant \sigma \leqslant 1),$$

where $c(\sigma)$ *is a constant depending only on* σ.

Theorem 3.

(5.3)

$$\int_0^T \mathrm{I} f(\sigma+it)\,dt = \int_\sigma^\infty \log|\zeta(\sigma+iT)|\,d\sigma + c'(\sigma) \quad (-1 \leqslant \sigma \leqslant 2),$$

where $c'(\sigma)$ *is a constant depending only on* σ. *In particular*

$$\int_0^T S(t)\,dt = \frac{1}{\pi}\int_{\frac{1}{2}}^\infty \log|\zeta(\sigma+iT)|\,d\sigma + c_1 \tag{5.4}$$

$$= \frac{1}{\pi}\int_{\frac{1}{2}}^2 \log|\zeta(\sigma+iT)|\,d\sigma + O(1). \tag{5.5}$$

* It is easily verified that $\int d\sigma \int |\phi'/\phi|\,dt$ exists: in fact near a singularity σ_0+it_0 of ϕ'/ϕ we have

$$\int |\phi'/\phi|\,dt = O(\log|\sigma-\sigma_0|)$$

and the integral of this with respect to σ converges. We may invert the order of integration and ignore lines on which the inner integrals do not exist.

Theorems 2 and 3 are simple consequences of Theorem 1 and known results. We take in Theorem 1 $\alpha = \sigma$, β large and positive, $\phi(s) = \zeta(s)$. Then (on account of the pole at $s = 1$)

$$\nu(\sigma, T) = \begin{cases} N(\sigma, T) - \frac{1}{2} & (\sigma < 1) \\ 0 & (\sigma \geqslant 1). \end{cases}$$

Theorem 1 gives

$$-2\pi i \int_\sigma^1 \nu(\sigma, T)\, d\sigma = -2\pi i \int_\sigma^\beta \nu(\sigma, T)\, d\sigma$$

$$= \int_\sigma^\beta f(\sigma)\, d\sigma - \int_\sigma^\beta f(\sigma + iT)\, d\sigma - \int_0^T f(\sigma + it)\, i dt + \int_0^T f(\beta + it)\, i dt.$$

If in this we make $\beta \to \infty$ the last term disappears, and the first two integrals tend to integrals $\int_\sigma^\infty$. Taking real and imaginary parts we obtain (5.1) and (5.3), and observe further that c_σ, c_σ' are of the form $O(1)$ for $\sigma \geqslant -1$. (5.2) now follows from (5.1) in virtue of (1.4)*, (5.4) is the special case $\sigma = \frac{1}{2}$ of (5.3), and (5.5) follows at once.

6. We prove next the following results.

Theorem 4.

(6.1)

$$\int_\sigma^1 N(\sigma, T)\, dt = O(T)\, \mathrm{Min}\left(\log\log T, \log\frac{1}{\sigma - \frac{1}{2}}\right) \quad (\tfrac{1}{2} \leqslant \sigma \leqslant 1).$$

In particular

$$\int_{\frac{1}{2}}^1 N(\sigma, T)\, d\sigma = O(T \log\log T). \tag{6.2}$$

Theorem 5.

$$N(\sigma, T) = O\left(T \frac{1}{\sigma - \frac{1}{2}} \log\frac{1}{\sigma - \frac{1}{2}}\right) \quad (\tfrac{1}{2} < \sigma \leqslant 1). \tag{6.3}$$

From (5.1)

$$2\pi \int_\sigma^1 N(\sigma, T)\, d\sigma = \int_2^T \log|\zeta(\sigma + it)|\, dt + O(\log T)$$

$$= \tfrac{1}{2} \int_2^T \log|\zeta(\sigma + it)|^2\, dt + O(\log T)$$

$$\leqslant \tfrac{1}{2}(T - 2) \log\left\{\int_2^T |\zeta(\sigma + it)|^2\, dt \Big/ (T - 2)\right\} + O(\log T)$$

$$< AT \log\left\{\mathrm{Min}\left[O(\log T),\ O\left(\log\frac{1}{\sigma - \frac{1}{2}}\right)\right]\right\} + O(\log T),$$

* Since $f = O(2^{-\sigma})$ $(\sigma \geqslant 2)$, so that $\int_2^\infty f\, d\sigma = O(1)$.

by Lemma 1; and this proves (6.1). We now observe that if $\sigma_1 = \frac{1}{2} + \frac{1}{2}(\sigma - \frac{1}{2})$,

$$N(\sigma, T) \leqslant \frac{1}{\sigma_1 - \sigma} \int_{\sigma_1}^{\sigma} N(\sigma, T)\, d\sigma \leqslant \frac{2}{\sigma - \frac{1}{2}} \int_{\sigma_1}^{1} N(\sigma, T)\, d\sigma$$

$$< AT(\sigma - \tfrac{1}{2})^{-1} \log(\sigma_1 - \tfrac{1}{2})^{-1} < AT(\sigma - \tfrac{1}{2})^{-1} \log(\sigma - \tfrac{1}{2})^{-1},$$

which is (6.3).

Theorem 6. *If $\phi(T)$ is positive, and increases to ∞ with T, then all but an infinitesimal proportion of the zeros of $\zeta(s)$ in the upper half plane lie in the region*

$$|\sigma - \tfrac{1}{2}| < \phi(t) \frac{\log\log t}{\log t}, \quad t > e^2. \tag{6.4}$$

That is to say: the number of zeros outside the region and with imaginary part between 0 and T is of the form $o(T \log T)$ (the total number between these limits being approximately $\frac{1}{2\pi} T \log T$). It is clearly enough to prove that for large T the number of zeros in the region

$$\sigma \geqslant \tfrac{1}{2} + \phi(t) \frac{\log\log t}{\log t}, \quad \sqrt{T} < t < T, \tag{6.5}$$

is $o(T \log T)$. The curved boundary of (6.5) lies to the right of $\sigma = \sigma_1$, where

$$\sigma_1 - \tfrac{1}{2} = \phi(\sqrt{T}) \frac{\log\log T}{\log T}.$$

But evidently $N(\sigma_1, T) = o(T \log T)$ in virtue of Theorem 5*, and the required result is true *a fortiori*.

It should be observed that Theorems 4, 5 and 6 are novel only so far as they concern a *small* neighbourhood of the line $\sigma = \frac{1}{2}$: for *fixed* $\sigma > \frac{1}{2}$ Carlson's well-known theorem† gives results much more precise than ours.

7. We turn now to the study of the function $S(T)$, or rather of its integral $\int_0^T S(t)\, dt$. We prove two theorems.

Theorem 7‡. $$\int_0^T S(t)\, dt = O(\log T).$$

* If $\sigma_1 > 1$ the result is trivial. If not, then $\sigma_1 - \frac{1}{2} \leqslant \frac{1}{2}$, and

$$\log \frac{1}{\sigma_1 - \frac{1}{2}} = O(\log\log T).$$

† F. Carlson, *Arkiv för Mat., Ast. och Fys.*, 15 (1920), No. 20. The details of the proof can be lightened by Theorem 2, which enables us to reduce the double integration to a single one. Carlson's theorem lies much deeper than ours for fixed σ, but it appears to throw no light on the present problem.

‡ H. Cramér, *Math. Zeitschr.*, 4 (1919), 104–130, 122–130, proves, by an entirely different method, that $\int_0^T S(t)\, dt = O(T^{\epsilon})$.

Theorem 8. *It follows from Lindelöf's hypothesis that*

$$\int_0^T S(t)\,dt = o(\log T).$$

We begin with the proof of the more difficult Theorem 8. Lindelöf's hypothesis is that

$$|\zeta(\tfrac{1}{2}+it)| < t^{o(1)},$$

and this is known* to involve the proposition that the number of zeros of $\zeta(s)$ in any region, D,

$$\sigma \geqslant \tfrac{1}{2}+\delta, \quad |t-T| \leqslant 1,$$

is of the form $o(\log T)$†. We suppose $\delta < \frac{1}{10}$. Let C_3 be the (boundary of the) rectangle with corners at $\frac{1}{2}+2\delta+i\{T\pm(1-\delta)\}$, $2\pm i\{T\pm(1-\delta)\}$; C_2 that with corners at

$$\tfrac{1}{2}+3\delta+i\{T\pm(1-2\delta)\},\quad 2-\delta+i\{T\pm(1-2\delta)\};$$

C_1 that with corners at $\frac{3}{2}+i\{T\pm(1-3\delta)\}$, $\frac{5}{3}+i\{T\pm(1-3\delta)\}$. We denote by Σ_0 a summation over the zeros ρ of $\zeta(s)$ for which $|\gamma-T| \leqslant 1$, Σ_1 a summation over those zeros lying in D (the terms of Σ_1 are therefore contained in Σ_0). It is known that Σ_0 has $O(\log T)$ terms‡. Let now

$$\psi(s) = \psi(s, T, \delta) = \frac{\zeta'(s)}{\zeta(s)} - \Sigma_1 \frac{1}{s-\rho},$$

so that $\psi(s)$ is regular in C_3.

8. *Lemma* 3. *bis* $\quad \psi(s) = O\left(\dfrac{\log T}{\delta}\right)$ (*uniformly*) *for* s *in* C_3.

We have, for s in C_3,

$$\frac{\zeta'(s)}{\zeta(s)} = O(\log t) + \Sigma\left(\frac{1}{s-\rho}+\frac{1}{\rho}\right)\text{§},$$

$$\psi(s) = O(\log t) + (\Sigma-\Sigma_1)\left(\frac{1}{s-\rho}+\frac{1}{\rho}\right) + \Sigma_1\frac{1}{\rho} \tag{8.1}$$

$$= O(\log T) + (\Sigma-\Sigma_1)\left(\frac{1}{s-\rho}+\frac{1}{\rho}\right).$$

If Σ' denotes a summation over ρ for which $|\gamma-t| \geqslant 1$ we have‖

$$\Sigma'\left(\frac{1}{s-\rho}+\frac{1}{\rho}\right) = O(\log t) = O(\log T). \tag{8.2}$$

* R. Backlund, *Öfversigt Finska Vetensk. Soc.*, 61 (1918–19), No. 9.

† o's are not uniform in δ. O's (by our current convention) are absolute, and so uniform in δ. We must therefore bear in mind that, e.g., a $o(\log T)$ is not necessarily of the form $O(\log T)$; for example δ^{-1} is of the first form but not of the second.

‡ E. Landau, *Handbuch*, p. 337. We refer to this treatise as *H.B.*

§ *H.B.*, pp. 316–318. ‖ *H.B.*, p. 339.

Since the number of ρ within a vertical distance $O(1)$ of T is $O(\log T)$ we may write

$$\Sigma - \Sigma_1 = \Sigma' + \Sigma_2 - \Sigma_3,$$

where Σ_2 and Σ_3 are exclusive and contain $O(\log T)$ terms, and where further $|\, \mathrm{I}\!\mathrm{I}\, (s-\rho)\,| \geqslant 1$ in Σ_3, and in Σ_2

$$|\, s-\rho\,| \geqslant \textit{distance between } C_2 \textit{ and boundary of } D \geqslant \delta.$$

Hence

$$(\Sigma - \Sigma_1)\left(\frac{1}{s-\rho}+\frac{1}{\rho}\right) = \Sigma'\left(\frac{1}{s-\rho}+\frac{1}{\rho}\right) + \Sigma_2\, O\left(\frac{1}{\delta}\right) - \Sigma_3\, O(1)$$
$$= O\left(\frac{\log T}{\delta}\right)$$

by (8.2). The lemma now follows from (8.1).

9. *Lemma* 4.

(9.1) $\psi(s) = o(\log T) \quad (\frac{1}{2} + 3\delta \leqslant \sigma \leqslant \frac{5}{3},\ |\, t-T\,| \leqslant \frac{1}{2})$.

Consider $\psi(s)$ on C_1. Here $\sigma \geqslant \frac{3}{2}$ and so

$$\frac{\zeta'(s)}{\zeta(s)} = O(1).$$

Since $1/(s-\rho) = O(1)$ in Σ_1 and Σ_1 has $o(\log T)$ terms, we have $\psi(s) = o(\log T)$. Let now

$$M_1 = \operatorname*{Max}_{C_1} |\, \psi\,|, \quad M_3 = \operatorname*{Max}_{C_3} |\, \psi\,|.$$

C_1 is interior to C_2 and C_2 to C_3 (since $\delta < \frac{1}{10}$), and Lemma 3 is applicable to $\psi(s)$, the indices α, β depending only on δ. Thus in and upon C_2

$$|\, \psi(s)\,| \leqslant M_1^{\alpha} M_3^{\beta} < \{o(\log T)\}^{\alpha} \left\{O\left(\frac{\log T}{\delta}\right)\right\}^{\beta} = o(\log T).$$

But C_2 includes the region with which the lemma is concerned.

10. We suppose now that neither t nor T is the ordinate of a ρ. Then we may integrate (9.1) between s and $\frac{5}{3}+it$, and obtain (all logarithms having their principal values)

$$f(s) - f(\tfrac{5}{3}+it) = \Sigma_1 \{\log(s-\rho) - \log(\tfrac{5}{3}+it-\rho)\} + \int_{\frac{5}{3}+it}^{s} \psi(s)\, ds,$$

$$(10.1)\quad f(s) = O(1) + \Sigma_1 \log(s-\rho) - \Sigma_1\, O(1) + o(\log T)$$
$$= o(\log T) + \Sigma_1 \log(s-\rho) \quad (\tfrac{1}{2} + 3\delta \leqslant \sigma \leqslant \tfrac{5}{3},\ |\, t-T\,| \leqslant \tfrac{1}{2}),$$

since Σ_1 has $o(\log T)$ terms. Integrating again between the same limits (s always satisfying the conditions of (10.1))

$$\int_{\sigma}^{\frac{5}{3}} f(\sigma+it)\, d\sigma = o(\log T) + \Sigma_1 \int_{\sigma}^{\frac{5}{3}} \log(s-\rho)\, d\sigma.$$

It is clear (and follows strictly from Lemma 2 with $s-\rho=z, \eta=2$) that the integral last written is $O(1)$. Therefore

(10.2)
$$\int_\sigma^2 f(\sigma+it)\,d\sigma = O(1)+\int_\sigma^{\frac{5}{3}} f(\sigma+it)\,d\sigma = o(\log T)+\Sigma_1 O(1) = o(\log T).$$

From this follows

Lemma 5. *There exists a* σ_0 *satisfying* $\frac{1}{2}+3\delta \leqslant \sigma_0 \leqslant \frac{1}{2}+4\delta$ *and such that*

$$\log|\zeta(s_0)| = o(\log T) \quad (s_0=\sigma_0+iT), \tag{10.3}$$

$$\int_{\sigma_0}^2 \log|\zeta(\sigma+iT)|\,d\sigma = o(\log T), \tag{10.4}$$

$$\Sigma_1 \log|s_0-\rho| = o(\log T). \tag{10.5}$$

In fact (10.4) is true, by (10.2), for any $\sigma_0 \geqslant \frac{1}{2}+3\delta$. Also, by the same equation,

$$\begin{aligned}\left|\frac{1}{\delta}\int_{\frac{1}{2}+3\delta}^{\frac{1}{2}+4\delta}\log|\zeta(\sigma+iT)|\,d\sigma\right| &\leqslant \frac{1}{\delta}\left|\int_{\frac{1}{2}+3\delta}^{\frac{1}{2}+4\delta} f(s)\,d\sigma\right| \\ &\leqslant \frac{1}{\delta}\left|\int_{\frac{1}{2}+3\delta}^{2} f(s)\,d\sigma\right| + \frac{1}{\delta}\left|\int_{\frac{1}{2}+4\delta}^{2} f(s)\,d\sigma\right| \\ &< o(\log T)+o(\log T) = o(\log T).\end{aligned}$$

Since the average of $\log|\zeta|$ over $\sigma=\frac{1}{2}+3\delta$ to $\frac{1}{2}+4\delta$ has a modulus $o(\log T)$ (and $\log|\zeta|$ is continuous), the value of $\log|\zeta|$ itself must be $o(\log T)$ for some $\sigma=\sigma_0$. Finally (10.5) follows from (10.1) and (10.3).

11. For $s=\sigma+iT$ we have (since $\Sigma=\Sigma_0+\Sigma'$ when $t=T$)

$$\begin{aligned}&\frac{\zeta'(s)}{\zeta(s)} - \Sigma_0\frac{1}{s-\rho} \\ &= O(\log T)+\Sigma'\left(\frac{1}{s-\rho}+\frac{1}{\rho}\right)+\Sigma_0\frac{1}{\rho} = O(\log T) \quad (\sigma\geqslant\tfrac{1}{2}).\end{aligned} \tag{11.1}$$

Still supposing only $\sigma \geqslant \frac{1}{2}$, integrating between s and s_0, and taking real parts, we obtain

$$\begin{aligned}\log|\zeta(s)| - \log|\zeta(s_0)| &= \Re\,\Sigma_0\int_{s_0}^{s}\frac{ds}{s-\rho}+O(\log T) \\ &= \Sigma_0\log(s-\rho)-\Sigma_0\log(s_0-\rho)+O(\log T),\end{aligned}$$

and so

$$\begin{aligned}\log|\zeta(s)| &= O(\log T)+o(\log T) \\ &\quad +\Sigma_0\log|s-\rho|-(\Sigma_0-\Sigma_1)\log|s_0-\rho|,\end{aligned} \tag{11.2}$$

since $\log|\zeta(s_0)|$ and $\Sigma_1\log|s_0-\rho|$ are $o(\log T)$.

Now for a term of $\Sigma_0 - \Sigma_1$ we have $\mathfrak{R}\rho \leqslant \frac{1}{2} + \delta$, and so

$$\mathfrak{R}\,(s_0 - \rho) \geqslant 2\delta,$$

and

$$\log |\, s_0 - \rho \,| = O\left(\log \frac{1}{\delta}\right).$$

Hence, integrating (11.2) from $\sigma = \frac{1}{2}$ to $\sigma = \sigma_0$ (a distance not exceeding 4δ), we have

$$(11.3)\quad \int_{\frac{1}{2}}^{\sigma_0} \log |\, \zeta(s) \,|\, d\sigma = O(\delta \log T) + o(\log T) + \Sigma_0\, \mathfrak{R} \int_{\frac{1}{2}}^{\sigma_0} \log (s - \rho)\, d\sigma + (\Sigma_0 - \Sigma_1)\, O\left(\delta \log \frac{1}{\delta}\right).$$

Now in Σ_0 $\quad \mathfrak{R} \int_{\frac{1}{2}}^{\sigma_0} \log (s - \rho)\, d\sigma = \mathfrak{R} \int_{z_1}^{z_2} \log z\, dz,$

where z_1, z_2 satisfy the conditions of Lemma 2, with $\eta = 4\delta$. Hence

$$(11.4)\quad \int_{\frac{1}{2}}^{\sigma_0} \log |\, s - \rho \,|\, d\sigma = O\left(\delta \log \frac{1}{\delta}\right).$$

Since Σ_0 and $\Sigma_0 - \Sigma_1$ have $O(\log T)$ terms we have finally, from (11.3) and (11.4),

$$(11.5)\quad \int_{\frac{1}{2}}^{\sigma_0} \log |\, \zeta(s) \,|\, d\sigma = O\left(\delta \log \frac{1}{\delta} \log T\right) + o(\log T).$$

Combining (11.5) and (10.4), we obtain

$$\int_{\frac{1}{2}}^{2} \log |\, \zeta(s) \,|\, d\sigma = o(\log T) + O\left(\delta \log \frac{1}{\delta} \log T\right),$$

and so, since the left side is independent of δ, and $\delta \log 1/\delta \to 0$ with δ,

$$(11.6)\quad \int_{\frac{1}{2}}^{2} \log |\, \zeta(s) \,|\, d\sigma = o(\log T).$$

It follows from Theorem 3 that $\int_0^T S(t)\, dt = o(\log T)$, when, as we have supposed, T is not the ordinate of a ρ. But the result, if true for such T, is true for all, and the proof of Theorem 8 is completed.

12. The proof of Theorem 7 follows similar, but much simpler lines. Let

$$\chi(s) = \chi(s, T) = \frac{\zeta'(s)}{\zeta(s)} - \Sigma_0 \frac{1}{s - \rho}.$$

We suppose $t = T$, or $s = \sigma + iT$, throughout. Then (by (8.2))

$$\chi(s) = O(\log T).$$

Integration gives

$$f(s) - \Sigma_0 \log (s - \rho) = f(2 + iT) - \Sigma_0 \log (2 + iT - \rho) + O(\log T) = O(1) + \Sigma_0\, O(1) + O(\log T) = O(\log T).$$

Integrating again and taking real parts we obtain

$$\int_{\frac{1}{2}}^{2} \log|\zeta(\sigma+iT)|\,d\sigma = \Re\int_{\frac{1}{2}}^{2} f(s)\,d\sigma = O(\log T)+\Sigma_0\,\Re\int_{\frac{1}{2}}^{2}\log(s-\rho)\,d\sigma$$
$$= O(\log T)+\Sigma_0\,\Re\,O(1) = O(\log T).$$

Theorem 7 follows from this and (5.5).

13. *Theorem* 9. *It follows from Lindelöf's hypothesis that*

$$S(T) = o(\log T).$$

This is a known theorem, due to Cramér*. It is, however, interesting to observe that it is an almost immediate consequence of Theorem 8. Suppose the result false. Then there exists a positive h such that either $S(T) > h\log T$ for arbitrarily large T, or else $S(T) < -h\log T$ for arbitrarily large T (both events, of course, can occur). We observe that

(13.1) $S(T+x)-S(T)$
$= \{N(T+x)-N(T)\}-\{M(T+x)-M(T)\}+O(1)$
$\geqslant -M(T+x)+M(T)-A$
$\geqslant -A_1 x\log T - A_2 \quad (T>2,\ 0\leqslant x\leqslant T),$

and similarly

(13.2) $S(T-x)-S(T)\leqslant A_3 x\log T + A_4 \quad (T>2,\ 0\leqslant x\leqslant T).$

If now $S(T) > h\log T$ for arbitrarily large T we have, writing $\xi = h/A_1$,

$$\int_0^{T+\xi} S(t)\,dt - \int_0^T S(t)\,dt = \int_0^\xi S(T+x)\,dx$$
$$= \xi S(T)+\int_0^\xi \{S(T+x)-S(T)\}\,dx$$
$$\geqslant \xi h\log T+\int_0^\xi(-A_1 x\log T - A_2)\,dx$$
$$= (\xi h-\tfrac{1}{2}A_1\xi^2)\log T + O(1) = \frac{h^2}{2A_1}\log T + O(1),$$

which is false, since the left side is $o(\log T)$. A similar proof disposes of the other case: we consider $\int_0^{\xi'} S(T-x)\,dx$ and use (13.2).

14. We now enter upon the second part of the paper. We always assume the Riemann hypothesis, and do not always state it explicitly in enunciating our theorems and lemmas.

Let $S_0(T) = S(T)$,

$$S_{2n}(T) = \frac{(-1)^n}{\pi}\,\Re\int_{\frac{1}{2}}^{\infty}\int_\sigma^\infty\int_\sigma^\infty \ldots f(\sigma+iT)\,(d\sigma)^{2n} \quad (n\geqslant 1),$$

$$S_{2n-1}(T) = \frac{(-1)^{n-1}}{\pi}\,\Re\int_{\frac{1}{2}}^{\infty}\int_\sigma^\infty\int_\sigma^\infty \ldots f(\sigma+iT)\,(d\sigma)^{2n-1} \quad (n\geqslant 1).$$

* *Math. Zeitschr.*, 2 (1918), 237–241.

Theorem 10.

$$S_n(T) = \int_0^T S_{n-1}(t)\,dt + c_n \quad (n > 0), \tag{14.1}$$

where c_n is a constant depending only on n.

This is deducible from Theorems 2 and 3 by repeated integration, if we observe that $N(\sigma, T) = 0$ $(\sigma \geqslant \frac{1}{2})$. It is, however, unnecessary to use these theorems. In fact, since $\log \zeta$ is regular in $\sigma > \frac{1}{2}$ (except at $s = 1$) we have, for $\sigma > \frac{1}{2}$, and by continuity for $\sigma \geqslant \frac{1}{2}$,

$$i\int_0^T f(\sigma + it)\,dt + \int_\sigma^\infty f(\sigma + iT)\,d\sigma = \int_\sigma^\infty f(s)\,ds,$$

the path of the last integral lying in $t > 0$. Deforming this path into the real axis, and taking account of the discontinuity of $f(s)$ along the whole stretch $\frac{1}{2} \leqslant \sigma \leqslant 1$, $t = 0$, we have, supposing $T > 0$, $\sigma \geqslant \frac{1}{2}$,

$$i\int_0^T f(\sigma + it)\,dt + \int_\sigma^\infty f(\sigma + iT)\,d\sigma = \int_\sigma^\infty f(s)\,ds + iC_\sigma' = C_\sigma + iC_\sigma', \tag{14.2}$$

where C_σ, C_σ' depend only on σ, $C_\sigma' = 0$ for $\sigma > 1$, and $C_\sigma = O(2^{-\sigma})$ as $\sigma \to +\infty$. Taking real and imaginary parts in (14.2) we obtain

$$\int_0^T \mathrm{I}f(\sigma + it)\,dt = \int_\sigma^\infty \log|\zeta(\sigma + iT)|\,d\sigma - C_\sigma', \tag{14.3}$$

$$\int_0^T \log|\zeta(\sigma + it)|\,dt = -\int_\sigma^\infty \mathrm{I}f(\sigma + iT)\,d\sigma + C_\sigma. \tag{14.4}$$

If we write $\sigma = \frac{1}{2}$ in (14.3) we obtain (14.1) with $n = 1$. A second integration gives, by means of (14.4),

$$\begin{aligned}\int_0^T S_1(t)\,dt &= \int_{\frac{1}{2}}^\infty d\sigma \int_0^T \log|\zeta(\sigma + it)|\,dt \\ &= -\int_{\frac{1}{2}}^\infty d\sigma \int_\sigma^\infty \{\mathrm{I}f(\sigma + iT)\,d\sigma + C_\sigma\}\,d\sigma \\ &= -\int_{\frac{1}{2}}^\infty d\sigma \int_\sigma^\infty \mathrm{I}f(\sigma + iT)\,d\sigma + c_2.\end{aligned}$$

The process may be repeated indefinitely, (14.3) and (14.4) being used alternately.

15. *Theorem* 11. *On the Riemann hypothesis*

$$S(T) = O\left(\frac{\log T}{\log\log T}\right)^*,$$

and, more generally,

$$S_n(T) = O\left(\frac{\log T}{(\log\log T)^{n+1}}\right) \quad (n \geqslant 0).$$

* Results intermediate between this and the $S(T) = O(\log T)$ which is true without the Riemann hypothesis have been given by Bohr, Cramér, and Landau.

The proof of this theorem is long and difficult, and depends on a singularly varied set of ideas. I begin with some preliminary lemmas.

Throughout the sequel $T > 4$. We shall adopt the abbreviations

$$l(t) = \log t, \quad \lambda(t) = \log\log t, \quad \mu(t) = 1/\lambda(t);$$

$$l = l(T) = \log T, \quad \lambda = \lambda(T) = \log\log T, \quad \mu = \mu(T) = 1/\lambda(T).$$

Thus we drop the argument T in l, λ, μ, but never the argument t.

Lemma 6 (i). *Let $f(T)$ be an increasing function, $f(T) > 1$, and $f(T) = O(l^2)$. Then for large T there exists a t between T and $T + l - 2$ for which $f(t+2) < \frac{3}{2} f(t)$.*

(ii) *Let f_1, f_2 be increasing functions of T, greater than unity, and of the form $O(l^2)$. Let $T_1(T) = O(l)$, $T_2(T) = O(l)$. Then for large T there exists a t between T and $T + l^4$ for which (simultaneously)*

$$f_1\{t + T_1(t)\} < \tfrac{3}{2} f_1(t), \tag{15.1}$$

$$f_2\{t + T_2(t)\} < \tfrac{3}{2} f_2(t). \tag{15.2}$$

If (i) is false we have, for some large T,

$$f(T + 2n) \geqslant \tfrac{3}{2} f(T + 2n - 2) \quad (0 < n \leqslant \tfrac{1}{2}l - 1).$$

Hence for some large T and $\nu = [\frac{1}{2}l] - 1$

$$f(2T) \geqslant f(T + 2\nu) \geqslant \tfrac{3}{2} f(T + 2\nu - 2) \geqslant \ldots \geqslant (\tfrac{3}{2})^{\nu} f(T) > (\tfrac{3}{2})^{\nu} > l^3(2T),$$

which is false.

Consider now (ii). Let $T_n = T_{n-1} + l^2(T_{n-1})$ $(n > 0)$, $T_0 = T$. Then $T_n - T_{n-1} < l^2(T_n)$, and if $n \leqslant l$

$$T_n - T < n l^2(T_n) < l^3(T_n),$$

from which it follows (for large T) that $T_n < 2T$, and so, for $n \leqslant N = [l]$,

$$T_n - T < l^4. \tag{15.3}$$

Now the pair of inequalities

$$f_1(T_n) < \tfrac{3}{2} f_1(T_{n-1}) \tag{15.4}$$

$$f_2(T_n) < \tfrac{3}{2} f_2(T_{n-1}) \tag{15.5}$$

$(n > 0)$

must hold simultaneously for some $n \leqslant N$. For if not, one of them, say the first, must be false for at least $\frac{1}{2}N$ values of n, from which it follows that

$$f_1(T_N) \geqslant (\tfrac{3}{2})^{\frac{1}{2}N} f_1(T) > e^{AN}$$

and so

$$f_1(T + l^4) \geqslant f_1(T_N) > l^3,$$

which is false. But for an n giving (15.4) and (15.5) we have, taking $t = T_{n-1}$,

$$f_1\{t + T_1(t)\} \leqslant f_1\{t + l^2(t)\} = f_1(T_n) < \tfrac{3}{2} f_1(t)$$

for large T. This is (15.1), and (15.2) follows similarly.

16. *Lemma* 7. *Suppose* $R(T)$ *is continuous in* $T > 0$ *except at isolated points, and*

$$R_1(T) = \int_0^T R(t)\,dt + b_1, \quad R_2(T) = \int_0^T R_1(t)\,dt + b_2,$$

where b_1, b_2 *are constants. Suppose further that as* $T \to \infty$

$$R(T) = O\{g(T)\,h(T)\},$$

$$R_2(T) = O\{g(T)\}^*,$$

where $g(T)$ *is increasing and greater than unity, and* $T^2h(T)$ *is increasing and greater than unity, for* $T > T_0$. *Then*

$$R_1(T) = O\{g(T)\sqrt{h(T)}\}.$$

This is practically included in a known result†.

We require also another lemma, belonging to the same order of ideas, and embodying an argument of which we have had an example already in § 13.

Lemma 8. *Suppose* $\psi(T) = O(l)$ *is an increasing function greater than unity* $(T > 4)$, *and*

$$R_1(T) = \int_0^T R(t)\,dt + O(1).$$

Suppose further that

$$R(T+x) - R(T) > -Kxl - K' \quad (0 \leqslant x < \tfrac{1}{2}T), \tag{16.1}$$

where K *and* K' *are constants, and that*

$$R_1(T) = O\{l\mu^2\psi(T)\}. \tag{16.2}$$

Then

$$R(T) = O(l\mu)\sqrt{\psi(2T)}. \tag{16.3}$$

We write, for brevity, $u = \sqrt{\psi(2T)}$. If (16.3) is false, then, for any positive h (however large), there are arbitrarily large T for which either

$$R(T) > hl\mu u, \tag{16.4}$$

or else

$$R(T) < -hl\mu u. \tag{16.5}$$

We suppose $h > K + 1$ (as we may), and choose $\xi = x = \mu u$, or

* The O's here are naturally not absolute.

† See G. H. Hardy and J. E. Littlewood, *Proc. London Math. Soc.* (2), 11 (1913). If $R(T)$ is continuous the theorem is a case of Theorem 6 (a), with

$$r=2, \quad x=T, \quad f(x)=R_2(T), \quad \phi(x)=g(T), \quad \psi(x)=T^2g(T)\,h(T).$$

In the actual case we can modify $R(T)$ in a small neighbourhood of each discontinuity so as to make it continuous, altering $R_1(T)$, $R_2(T)$ by arbitrarily little.

$\xi = -x$, according as (16.4) or (16.5) holds. Then $x < \frac{1}{2}T$ for large T, and we have respectively in the two cases

$$\int_T^{T+\xi} R(t)\,dt = \int_0^x R(T+y)\,dy = xR(T) + \int_0^x \{R(T+y) - R(T)\}\,dy$$
$$\geqslant xR(T) - \int_0^x (Ky\log T + K')\,dy,$$

and

$$\int_T^{T+\xi} R(t)\,dt = -\int_0^x R(T-y)\,dy = -xR(T) + \int_0^x \{R(T) - R(T-y)\}\,dy$$
$$\geqslant -xR(T) - \int_0^x \{Ky\log(T-y) + K'\}\,dy.$$

In either case

$$(16.6)\quad \left|\int_T^{T+\xi} R(t)\,dt\right| \geqslant x\,|\,R(T)\,| - \int_0^x (Ky\log T + K')\,dy$$
$$\geqslant xhl\mu u - \tfrac{1}{2}Klx^2 - K'x = x(h - \tfrac{1}{2}K)\,l\mu u - K'x$$
$$\geqslant x(\tfrac{1}{2}hl\mu u - K') > x(\tfrac{1}{4}hl\mu u) = \tfrac{1}{4}hl\mu^2\psi(2T).$$

But the left side does not exceed

$$|\,R_1(T+\xi)\,| + |\,R_1(T)\,| + O(1)$$
$$= O\left(\frac{\log(\frac{3}{2}T)}{\lambda(\frac{1}{2}T)}\,\psi(\tfrac{3}{2}T)\right) + O\{l\mu\psi(T)\} = O(l\mu)\,\psi(2T),$$

and it is impossible for (16.6) to hold for arbitrarily large h.

17. Let

$$P(T) = \max_{4\leqslant t\leqslant T}(\pi\,|\,S(t)\,|) + l\mu + 1, \qquad \bar{P}(T) = \max_{\substack{4\leqslant t\leqslant T\\ \sigma\geqslant\frac{1}{2}}}(|\,\mathbb{I}f(s)\,|) + l\mu + 1,$$

$$Q(T) = \max_{4\leqslant t\leqslant T}(|f(\tfrac{1}{2}+\mu+iT)|) + l\mu + 1, \quad \bar{Q}(T) = \max_{\substack{4\leqslant t\leqslant T\\ \sigma\geqslant\frac{1}{2}+\mu}}(|f(s)|) + l\mu + 1.$$

A word or two on the relations of these four functions may make them more easily distinguishable. Since $S(t)$ is substantially $\mathbb{I}f(s)$ with $\sigma = \frac{1}{2}$ we have, substantially, $P(t) \leqslant \bar{P}(t)$; and similarly $Q(t) \leqslant \bar{Q}(t)$. The Q's differ from the corresponding P's (1) by the presence of f instead of $\mathbb{I}f$; and (2) by the addition of μ to σ or the lower bound of σ (f, but not $\mathbb{I}f$, has infinities when $\sigma = \frac{1}{2}$, and σ must be kept from too near an approach to $\frac{1}{2}$). $\lambda\mu + 1$ is added to all the maxima to secure that the four functions exceed unity and are of order at least $l\mu$. Finally we take maxima over $4 \leqslant t \leqslant T$ to make our functions increasing. It will appear eventually that all four functions are of the same order, viz. $O(l\mu)$.

Lemma 9. *We have*

$$\bar{P}(T) = O(l), \quad \bar{Q}(T) = O(l\lambda).$$

The first of these follows from (1.4). To prove the second we observe that on the circle with centre at $s_0 = \frac{3}{2} + it_0$ ($t_0 \geqslant 4$) and radius $r = 1$,

$$|\mathfrak{R} f(s)| < A \log t_0.$$

Since $|f(s_0)| < A$ we have by Carathéodory's theorem

$$|f(s)| < \{A + Al(t_0)\} \frac{2r}{r-\rho}$$

in and upon the concentric circle of radius $\rho < r$. Taking

$$\rho \leqslant r - \mu = 1 - \mu,$$

we have, in particular,

$$|f(\sigma + it_0)| < Al(t_0)\lambda \quad (\tfrac{1}{2} + \mu \leqslant \sigma \leqslant \tfrac{3}{2}).$$

We may clearly omit the restriction $\sigma \leqslant \frac{3}{2}$, and the second result of the lemma follows by taking the maximum in the range $4 \leqslant t_0 \leqslant T$.

18. *Lemma* 10. *We have*

$$\bar{P}(T) = O\{P(T + l)\},$$
$$\bar{Q}(T) = O\{Q(T + l)\}.$$

In proving this we use the method of "rotation of domains" introduced by Lindelöf*. Consider the functions

$$g(s) = e^{\pm if(s)}$$

in the strip bounded by the vertical lines $\sigma = \frac{1}{2}$, $\sigma = \frac{3}{2}$, and the horizontal line $t = 4$. Let s_0 be a point of the strip for which $5 \leqslant t_0 \leqslant T$. Rotate the strip about the point s_0 through an angle $-\frac{1}{2}\pi$, and let $g_1(s)$ be the function that takes the value $g(s)$ at the point s_1 of the new strip corresponding to the point s of the original. Then s_0 is enclosed in a square, the length of whose side is unity, formed by the original vertical lines and their new positions. On the vertical boundaries of the original strip

$$|g(s)| \leqslant \exp|\mathfrak{R} f(s)|$$

does not exceed $\exp\{P(t)\}$ or $\exp A$ according as the boundary is the left or right hand one. In either case $|g(s)| < \exp\{P(t) + A\}$. For internal points of the strip we have $|g(s)| < \exp\{\bar{P}(t)\}$. It follows that for any point s of a vertical boundary of the square

$$|g(s)| < \exp\{P(T + 1) + A\}, \quad |g_1(s)| < \exp\{\bar{P}(T + 1)\},$$

and that for any point of a horizontal boundary of the square

$$|g(s)| < \exp\{\bar{P}(T + 1)\}, \quad |g_1(s)| < \exp\{P(T + 1) + A\},$$

* E. Lindelöf, *Acta Soc. Scient. Fenn.*, 46 (1915).

and so

$$(18.1)\quad |h(s)| = |g(s)g_1(s)| < \exp\{P(T+1) + \bar{P}(T+1) + A\}$$

on the whole boundary of the square. Since $h(s)$ is regular and bounded in the interior of the square, and continuous at the boundary except at a finite number of points, we conclude, by a well-known principle, that (18.1) is valid in the interior of the square, and in particular at $s = s_0$, where $g(s_0) = g_1(s_0)$. Thus

$$|g(s_0)|^2 < \exp\{P(T+1) + \bar{P}(T+1) + A\}.$$

Since one or other of the $\pm$ signs in the definition of g gives $|g| = \exp|\mathfrak{I}f|$ we conclude, dropping the suffix 0, that

$$2|\mathfrak{I}f(s)| \leqslant P(T+1) + \bar{P}(T+1) + A \quad (\tfrac{1}{2} < \sigma < \tfrac{3}{2},\ 5 \leqslant t \leqslant T).$$

In this we may further write $4 \leqslant t$ for $5 \leqslant t$, and suppress the condition $\sigma < \frac{3}{2}$. Next we may, by continuity, infer the inequality at $\sigma = \frac{1}{2}$ except when $\frac{1}{2} + it$ is a zero of $\zeta(s)$. Finally, since $\mathfrak{I}f(s) = \frac{1}{2}\{f(s+0i) + f(s-0i)\}$, we may infer the inequality also in this remaining case if we replace $T+1$ by $T+2$. Thus

$$2|\mathfrak{I}f(s)| \leqslant P(T+2) + \bar{P}(T+2) + A \quad (\sigma \geqslant \tfrac{1}{2},\ 4 \leqslant t \leqslant T).$$

Taking the maximum of this for the range of t we have now

$$2\bar{P}(T) \leqslant P(T+2) + \bar{P}(T+2) + A + 2l\mu + 2.$$

By Lemma 6 (i) there exists a $t = t(T)$ between T and $T + l - 2$ for which

$$\bar{P}(t+2) < \tfrac{3}{2}\bar{P}(T).$$

Then

$$2\bar{P}(t) \leqslant P(t+2) + \bar{P}(t+2) + Al\mu < P(t+2) + \tfrac{3}{2}\bar{P}(t) + Al\mu,$$

$$\tfrac{1}{2}\bar{P}(t) < P(t+2) + Al\mu,$$

$$\bar{P}(T) \leqslant \bar{P}(t) < 2P(t+2) + Al\mu \leqslant 2P(T+l) + Al\mu < AP(T+l),$$

which proves the first part.

The second part is similar and a little more direct. Arguing with the strip bounded by $\sigma = \frac{1}{2} + \mu$, $\sigma = \frac{3}{2}$, $t = 4$, and with f in place of g, we obtain

$$|f(s_0)|^2 \leqslant \{Q(T+1) + A\}\,\bar{Q}(T+1) < AQ(T+1)\,\bar{Q}(T+1).$$

This time there is continuity at the boundary; therefore, taking the square root and the maximum,

$$\bar{Q}(T) < A\{Q(T+1)\,\bar{Q}(T+1)\}^{\frac{1}{2}} + l\mu + 1 < A\{Q(T+1)\,\bar{Q}(T+1)\}^{\frac{1}{2}}.$$

By Lemma 6 (i), for some t between T and $T + l - 2$,

$$\bar{Q}(t+1) < \tfrac{3}{2}\bar{Q}(t),$$

and so

$$\bar{Q}(t) < A\{\bar{Q}(t)\,Q(t+1)\}^{\frac{1}{2}},$$

$$\bar{Q}(t) < AQ(t+1),$$

$$\bar{Q}(T) \leqslant \bar{Q}(t) < AQ(T+l-2+1) < AQ(T+l)$$

the desired result.

19. *Lemma* 11.

(19.1) $f(s) = O\{\bar{Q}(T+1)\,e^{-a_1(\sigma-\frac{1}{2})\lambda}\} = O\{Q(T+2l)\,e^{-a_1(\sigma-\frac{1}{2})\lambda}\}$

(uniformly) for $\quad \frac{1}{2}+\mu \leqslant \sigma \leqslant 2,\ t \leqslant T.$

The second part follows from Lemma 10 and the first part, which we therefore consider. We write $x = \sigma - \frac{1}{2}$. It is sufficient to prove (19.1) for $\mu \leqslant x \leqslant \frac{2}{3} - \mu$. Consider three circles with a common centre at $s_0 = \frac{3}{2} + \mu + it$ and radii $r_3 > r_2 > r_1$, going through the points $\frac{1}{2}+\mu+it$, $\frac{1}{2}+x+it$, $\frac{7}{6}+it$ respectively. Let M_1, M_2, M_3 be the upper bounds of $|f(s)|$ in the circles. Then by Hadamard's "three circles" theorem

$$M_2 \leqslant M_1^{\vartheta} M_3^{1-\vartheta},$$

where $\vartheta = \log\frac{r_3}{r_2}\Big/\log\frac{r_3}{r_1}$ (and $0<\vartheta<1$). Now for $\mu \leqslant x \leqslant 1$, and large T,

$$\vartheta = \log\{1-(x-\mu)\}^{-1}\Big/\log\frac{1}{\frac{1}{3}+\mu} \geqslant A(x-\mu)/A = A(x-\mu),$$

$$M_1 \leqslant f(\tfrac{7}{6}) = A,$$

$$M_3 \leqslant \operatorname*{Max}_{\substack{\sigma\geqslant\frac{1}{2}+\mu \\ 3\leqslant t\leqslant T+1}} |f(s)| < \bar{Q}(T+1) + A < A\bar{Q}(T+1).$$

Hence

$$M_2 < A\bar{Q}(T+1)\{\bar{Q}(T+1)\}^{-\vartheta} < A\bar{Q}(T+1)(l\mu)^{-\vartheta}$$
$$< A\bar{Q}(T+1)(l^{\frac{1}{2}})^{-A(x-\mu)} < A\bar{Q}(T+1)\,e^{-A\lambda(x-\mu)} < A\bar{Q}(T+1)\,e^{-A\lambda x},$$

the desired result.

20. *Lemma* 12. $\quad Q(T) = O\{P(T+l^5)\}.$

We have, from Lemmas 10 and 11,

(20.1) $\quad \mathfrak{F}f(s) = O\{P(T+2l)\} \quad (x \geqslant 0,\ t \leqslant T),$

(20.2) $\quad f(s) = O\{Q(T+2l)\,e^{-a_1\lambda x}\} \quad (\mu \leqslant x \leqslant \frac{3}{2},\ t \leqslant T).$

Let* $\quad \xi = 2\,\dfrac{1+a_1}{a_1}\,\mu \log\left(\dfrac{Q(T+2l)}{P(T+2l)} + e\right) > 2\mu,$

so that $\xi = o(1)$ (by Lemma 9), and $Q(T+2l)\,e^{-\frac{1}{2}a_1\lambda\xi} < P(T+2l)$. Then, for $\xi \leqslant x \leqslant \frac{3}{2}$, $t \leqslant T$, we have from (20.2)

(20.3) $\quad |f(s)| = O\{Q(T+2l)\,e^{-a_1\lambda x}\} = O\{Q(T+2l)\,e^{-\frac{1}{2}a_1\lambda(x+\xi)}\}$
$$= O\{P(T+2l)\,e^{-\frac{1}{2}a_1\lambda x}\}.$$

Consider now the range $0 < x \leqslant \xi$, and the circles with common centre at $s_0 = \frac{1}{2} + \xi + it$, and radii ξ and x. On the outer circle

(20.4) $\quad \mathfrak{F}f(s) = O\{P[T+1+2l(T+1)]\} = O\{P(T+3l)\}$

by (20.1). Also, by (20.3) with $x = \xi$,

(20.5) $\quad f(s_0) = O\{P(T+2l)\}.$

* This part of the argument is one of the main keys to the proof of Theorem 11.

By (20.4), (20.5) and Carathéodory's theorem we have, on the inner circle and in particular at $s = \frac{1}{2} + x + it$,

$$f(s) = O\left\{\frac{\xi}{x} P(T + 3l)\right\} \quad (0 < x \leqslant \xi). \tag{20.6}$$

Combining this with (20.3) we obtain

$$f(s) = O\{P(T + 2l)\, e^{-a_1 \lambda x}\} + O\left\{\frac{\xi}{x} P(T + 3l)\right\} \quad (x > 0). \tag{20.7}$$

In this we take $x = \mu$, vary t and take the maximum. This gives

$$Q(T) = O\{P(T + 2l)\} + O\{\xi\lambda P(T + 3l)\} + O(l\mu)$$
$$= O\{\xi\lambda P(T + 3l)\} = O\left\{P(T + 3l)\left[\log\left(\frac{Q(T + 2l)}{P(T + 2l)} + e\right)\right]\right\},$$

since $\xi\lambda > 2$, $P(T + 3l) \geqslant l\mu$. Hence

$$U(T) = \frac{Q(T)}{P(T + 3l)} < A \log\left[\frac{Q(T + 2l)}{P(T + 2l)} + e\right] \tag{20.8}$$
$$= A\left\{\log[U(T) + e] + \log\left[\frac{Q(T + 2l) + eP(T + 2l)}{Q(T) + eP(T + 3l)}\right] + \log\frac{P(T + 3l)}{P(T + 2l)}\right\}$$
$$< A \log[U(T) + e] + A \log\frac{Q(T + 2l)}{Q(T)} + A \log\frac{P(T + 3l)}{P(T)}.$$

Now by Lemma 6 (ii) there exists a t between T and $T + l^4$ for which

$$\frac{Q[t + 2l(t)]}{Q(t)} < A, \quad \frac{P[t + 3l(t)]}{P(t)} < A.$$

Then from (20.8)

$$U(t) < A \log[U(t) + e] + A,$$

from which it follows that $U(t) < A$, or

$$Q(t) < AP[t + 3l(t)].$$

Hence
$$Q(T) \leqslant Q(t) = O\{P(T + l^5)\},$$

the desired result.

Lemma 13.

$$\mathbb{I}f(s) = O\{P(T + l^6)\, e^{-a_1 \lambda x}\} \quad (0 \leqslant x \leqslant \tfrac{3}{2},\ t \leqslant T).$$

The result is true for $x \leqslant \mu$ by (20.1), and for $\mu \leqslant x \leqslant \frac{3}{2}$ by (20.2) and Lemma 12.

21. *Lemma* 14. $P(T) = O(l\mu)$.

We have

$$S_2(T) = -\frac{1}{\pi}\int_0^\infty dx \int_0^\infty \mathbb{I}f(\tfrac{1}{2} + x + iT)\, dx = -\frac{1}{\pi}\int_0^{\frac{3}{2}} dx \int_x^{\frac{3}{2}} \mathbb{I}f\, dx + O(1)$$
$$= O(\mu^2)\, P(T + l^6) + O(1) = O(\mu^2)\, P(T + l^6)$$

by Lemma 13. Also $S(T) = O\{P(T + l^6)\}$.

The conditions of Lemma 7 are fulfilled if $R(T) = S(T)$, $R_1(T) = S_1(T)$, $R_2(T) = S_2(T)$, and we conclude that

$$S_1(T) = O(\mu) P(T + l^6). \tag{21.1}$$

Further, if R and R_1 have their meanings above, and if

$$\psi(T) = \operatorname*{Max}_{4 \leqslant t \leqslant T} \left(\frac{\lambda(t)}{l(t)} P[t + l^6(t)] \right),$$

$\psi(T)$ is greater than unity, increasing, and of the form $O(\lambda) = O(l)$. Also, for $0 \leqslant x \leqslant \frac{1}{2}T$,

$$\begin{aligned} &R(T + x) - R(T) \\ &= \{N(T + x) - N(T)\} - \{M(T + x) - M(T)\} + O(1) \\ &> -\{M(T + x) - M(T)\} - A > -Ax \log T - A. \end{aligned}$$

The conditions of Lemma 8 are therefore satisfied, and we have

$$|S(T)| = |R(T)| = O(l\mu)\sqrt{\psi(2T)}.$$

Since the function on the right is increasing we may write t for T on the left and take the maximum for $t \leqslant T$, obtaining

$$P(T) = O(l\mu)\sqrt{\psi(2T)} + O(l\mu) = O(l\mu)\sqrt{\psi(2T)}.$$

Hence again, for $t \leqslant T$,

$$P[t + l^6(t)] = O\{l(t)\,\mu(t)\}\sqrt{\psi(3t)},$$

$$\frac{\lambda(t)}{l(t)} P[t + l^6(t)] = O\{\sqrt{\psi(3t)}\} = O\{\sqrt{\psi(3T)}\},$$

and taking the maximum on the left,

$$\psi(T) < A\sqrt{\psi(3T)}. \tag{21.2}$$

Now (21.2) and the evident inequality

$$\psi(T) < Al \tag{21.3}$$

together imply

$$\psi(T) < A. \tag{21.4}$$

For we must have

$$\psi(3t) < 3\psi(t) \tag{21.5}$$

for some arbitrarily large values of t; otherwise

$$\psi(4 \cdot 3^n) \geqslant 3^n \psi(4)$$

and $\psi(T) > AT$ for some arbitrarily large T, contrary to (21.3). But now (21.5) and (21.2) involve

$$\psi(t) < A\sqrt{\psi(t)}, \quad \psi(t) < A,$$

for arbitrarily large t, and so, since ψ is increasing, for all t. Hence, since

$$P(T) \leqslant P(T + l^6) \leqslant l\mu\psi(T),$$

we have finally

$$P(T) = O(l\mu).$$

22. Theorem 11 now follows without difficulty. We have at once

$$S(T) = O\{P(T)\} = O(l\mu)$$

the first of the desired results. Next we have

$$\begin{aligned} S_{2n}(T) &= \frac{(-1)^n}{\pi}\int_0^\infty dx\int_x^\infty dx \ldots \int_x^\infty \mathfrak{I} f(\tfrac{1}{2}+x+iT)\,dx \\ &= \frac{(-1)^n}{\pi}\int_0^{\frac{3}{2}} dx\int_x^{\frac{3}{2}} dx \ldots \int_x^{\frac{3}{2}} \mathfrak{I} f dx + O(1) \\ &= O\{\mu^{2n} P(T+l^6)\} = O(l\mu \,.\, \mu^{2n}) \end{aligned}$$

by Lemma 13, the desired result for S_n when n is even. Finally the results for odd n follow at once by Lemma 7 if we take $R = S_{2n-2}$, $R_1 = S_{2n-1}$, $R_2 = S_{2n}$.

23. Our material enables us to prove also

Theorem 12.

$$\zeta(\tfrac{1}{2}+iT) = O\left\{\exp\left(\frac{A\log T}{\log\log T}\right)\right\}.$$

In fact

$$\log\zeta(\tfrac{1}{2}+\mu+iT) = O\{Q(T)\} = O\{P(T+l^6)\} = O(l\mu).$$

Also*, for $s = \sigma + iT$,

$$\mathfrak{R}\frac{\zeta'(s)}{\zeta(s)} = O(1) - \tfrac{1}{2}\log T + \sum_\rho \mathfrak{R}\left(\frac{1}{s-\rho}+\frac{1}{\rho}\right) \quad (\tfrac{1}{2} < \sigma \leqslant 1).$$

Since every term in $\sum_\rho$ *is positive*

$$\mathfrak{R}\frac{\zeta'(s)}{\zeta(s)} > -\tfrac{1}{2}\log T - A,$$

$$\begin{aligned} \mathfrak{R}\log\zeta(\tfrac{1}{2}+iT) &= \mathfrak{R}\log\zeta(\tfrac{1}{2}+\mu+iT) - \int_{\frac{1}{2}}^{\frac{1}{2}+\mu} \mathfrak{R}\frac{\zeta'}{\zeta}(\sigma+iT)\,d\sigma \\ &\leqslant O(l\mu) + \mu(\tfrac{1}{2}\log T + A) = O(l\mu), \\ |\zeta(\tfrac{1}{2}+iT)| &\leqslant \exp\{O(l\mu)\}. \end{aligned}$$

24. We prove, in conclusion, another theorem about the orders of $S(T)$, $S_1(T)$, ... as functions of $\log T$. Let

$$\alpha_n = \overline{\lim_{T\to\infty}} \frac{\log|S_n(T)|}{\log\log T},$$

and $\alpha = \alpha_0$. That is, on the one hand

$$S(T) = O\{(\log T)^{\alpha+\epsilon}\},$$

and, on the other, this result becomes false if α is replaced by any smaller number; and similar relations hold for S_n and α_n. We may call α the "order" of $S(T)$. It is known that $\alpha > 0$; and it is in fact true that

* *H.B.*, p. 316.

$$\alpha \geqslant \tfrac{1}{2}, \quad \alpha_n \geqslant \tfrac{1}{2}; \tag{24.1}$$

but the proof of this would take us too far afield. We can prove, however, within our present circle of ideas:

Theorem 13. *The number α_n is a convex function of n, and*

$$1 \geqslant \alpha \geqslant \alpha_1 \geqslant \alpha_2 \geqslant \ldots.$$

It follows at once from Theorem 11 that $\alpha_n \leqslant 1$. Further, supposing $\alpha_n \geqslant 0$*, and taking $R(T) = S_n(T)$, $R_1 = S_{n+1}$, $R_2 = S_{n+2}$, we see that Lemma 7 is valid with

$$g(T) = (\log T)^{\alpha_{n+2}+\epsilon}, \quad g(T)\,h(T) = (\log T)^{\alpha_n+\epsilon}.$$

It therefore gives (O's depending on n and ϵ)

$$S_{n+1}(T) = O\{(\log T)^{\frac{1}{2}(\alpha_n+\alpha_{n+2})+\epsilon}\},$$

$$\alpha_{n+1} \leqslant \tfrac{1}{2}(\alpha_n + \alpha_{n+2}),$$

so that α_n is a convex function of n. Since the α's are bounded above it follows from the convexity that α_n never increases, and this completes the proof.

The upper bounds for the $S_n(T)$ given in Theorem 11 decrease by a factor $1/\log\log T$ for each addition of unity to n. Theorem 13 and the result $\alpha_n > 0$ together show that this factor cannot be replaced (throughout) by any constant power of $1/\log T$, however small.

* The modifications necessary if $\alpha_n < 0$ or $\alpha_{n+2} = -\infty$ can be supplied by the reader. Since actually $\alpha_n > 0$ it is not worth while to set out the details.